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**Minor Project Proposal Report
On
CUSTOM COMPACT ANTENNA DESIGN AND RSSI-BASED INDOOR
POSITIONING SYSTEM FOR GPS-DENIED ENVIRONMENTS USING ESP32**

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March, 2023

ABSTRACT

Indoor positioning in GPS-denied environments remains a critical challenge in modern communication and navigation systems. This project presents a low-cost, locally fabricable custom 2.4 GHz microstrip patch antenna on FR4 substrate and integrates it with a five-node ESP32 RSSI-based indoor localization system. The proposed antenna is designed using the transmission line model and simulated in MATLAB Antenna Toolbox, then fabricated through toner-transfer chemical etching and validated by VNA measurement and two-antenna RSSI pattern rotation. The custom patch antenna is mounted exclusively on the mobile node of an ESP-NOW peer-to-peer positioning network consisting of three fixed anchor nodes, one mobile node, and one gateway ESP32. The gateway aggregates RSSI data from all anchors and forwards it to a MATLAB base station that applies a path loss model, least-squares trilateration, and a Kalman filter to estimate position in real time. Validation is performed at twenty test positions using CDF-based statistical analysis. The system targets a mean positioning error below 1.2 metres — a significant improvement over the 2.0–2.5 m error observed with the built-in ESP32 trace antenna — at a total hardware cost below NPR 10,000 using components available in Kathmandu and Pokhara. The project is motivated by Nepal’s seismic risk and mountainous GPS shadow zones, and is architected to allow a future upgrade to CSI fingerprinting on ESP32-S3 without hardware changes

Keywords: Microstrip Patch Antenna, RSSI Trilateration, Indoor Positioning, ESP32, ESP-NOW, FR4, Kalman Filter, GPS-Denied

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1. LIST OF ABBREVIATIONS

CDF	Cumulative Distribution Function
CSI	Channel State Information
DGS	Defected Ground Structure
DXA	Device-independent Measurement Unit
ESP32	Microcontroller by Espressif Systems
ESP-NOW	Espressif Now - Connectionless Wi-Fi Protocol
FR4	Flame Retardant 4 - PCB Substrate Material
GPS	Global Positioning System
MATLAB	Matrix Laboratory
PIFA	Planar Inverted-F Antenna
RSSI	Received Signal Strength Indicator
SMA	SubMiniature version A - RF Connector
SPI	Serial Peripheral Interface
SOL	Short Open Load
TDOA	Time Difference of Arrival
UART	Universal Asynchronous Receiver Transmitter
VNA	Vector Network Analyzer
Wi-Fi	Wireless Fidelity

2. INTRODUCTION

2.1. Background

Indoor positioning has emerged as a critical challenge in modern communication and navigation systems. Unlike outdoor environments where GPS provides reliable location data, GPS signals are severely attenuated or completely unavailable in indoor settings such as buildings, warehouses, and underground structures. This problem is particularly relevant in Nepal, where mountainous terrain in regions such as Pokhara and the surrounding Himalayan valleys further creates outdoor GPS shadow zones, making alternative localization approaches a practical necessity for mobile nodes including autonomous robots, asset tracking systems, and sensor nodes.

Several real-world applications demand low-cost indoor positioning at zone-level accuracy of one to two metres. In warehouses, knowing which zone a trolley or tracked asset occupies is sufficient for inventory management, yet existing commercial solutions carry costs prohibitive for small businesses and student-built systems. In laboratory and campus environments, mobile robots and sensor nodes must know their approximate location to navigate between workstations and associate measurements with physical positions. All of these applications share a common requirement: a positioning system that is low-cost, self-contained, and replicable from locally available components — a need especially pressing in Nepal where commercial solutions remain financially inaccessible for most student projects.

Radio Frequency (RF) based positioning using Received Signal Strength Indicator (RSSI) trilateration offers a low-cost, infrastructure-light alternative to GPS for indoor environments. Fixed anchor nodes measure the signal strength received from a mobile node and use the path loss model to estimate distance, from which the node's position is computed using trilateration. The accuracy of this system is directly and critically dependent on the radiation characteristics of the antenna mounted on the mobile node.

The system architecture uses five ESP32 modules: three fixed anchor nodes, one mobile node carrying the custom compact patch antenna, and one dedicated gateway ESP32. The gateway aggregates RSSI data wirelessly from all anchors via the ESP-NOW protocol and forwards the combined data to a laptop over a single USB-serial cable, reducing cable clutter and improving real-time performance. The current implementation employs RSSI-based trilateration; the system architecture is intentionally designed to permit a future upgrade to Channel State Information (CSI) fingerprinting without any hardware changes.

2.2. Problem statement

GPS-denied environments such as building interiors, hospital corridors, university laboratories, and terrain-blocked zones present a critical navigation challenge for mobile nodes. Existing low-cost RF positioning systems use generic built-in antennas whose radiation patterns are unknown in the deployment context, resulting in inconsistent RSSI measurements and positioning errors of 2–4 metres.

There is no low-cost, locally fabricable compact antenna solution designed specifically for RSSI-based indoor positioning in the Nepal context. The specific issues arising from the built-in ESP32 PCB trace antenna are:

1. Asymmetric and highly directional radiation pattern with deep nulls: when the mobile node rotates, anchors in null directions receive dramatically weaker RSSI even at the same physical distance, producing false range estimates of up to 1.5 metres of error.
2. No dedicated ground plane: the built-in antenna relies on the PCB copper pour, which is disturbed when the board is mounted on a metal chassis or near a battery, detuning the resonant frequency unpredictably.
3. No published characterization: Espressif does not provide S_{11} , gain, impedance, or radiation pattern data for the built-in trace antenna, making it an undocumented radiator in an engineering sense.
4. Non-reproducibility: any two groups building the same system get different RSSI readings depending on board revision, chassis material, and node orientation.

2.3. Objectives

- To design, simulate, fabricate, and validate a compact 2.4 GHz patch antenna on FR4 substrate and demonstrate its positioning accuracy improvement over the built-in ESP32 antenna in a five-node ESP-NOW indoor localization system using trilateration and Kalman filtering.

2.4. Project Features

The proposed system's key features include:

- Custom Antenna Design: A rectangular microstrip patch antenna (Stage 1) with optional compact variants — meandered, DGS, or PIFA-type (Stage 2) — designed on FR4 substrate using closed-form transmission line model equations and validated by VNA
- RSSI-Based Indoor Positioning: Fixed anchor nodes measure signal strength from the mobile node and compute position using least-squares trilateration and a Kalman filter, achieving zone-level accuracy of 0.8–1.5 metres.
- Decentralized ESP-NOW Architecture: Connectionless peer-to-peer protocol eliminates single-AP association latency, enabling all three anchors to receive a broadcast simultaneously in under 10 ms.
- Locally Fabricable and Low Cost: Antenna fabricated using toner-transfer and ferric chloride etching from FR4 copper-clad available in Kathmandu and Pokhara; total hardware cost below NPR 10,000.
- Reproducible Engineering Contribution: Fully documented dimensions, S_{11} , impedance, gain, and radiation pattern make the antenna reusable by subsequent student projects.
- Upgradeable Architecture: The hardware, antenna, test layout, and gateway are designed to support a future upgrade to CSI fingerprinting on ESP32-S3 without any hardware changes.

2.5. Feasibility Analysis

This section studies whether the proposed custom antenna design and RSSI-based positioning system can be practically developed and validated. Feasibility is examined from technical, operational, and economic perspectives.

2.5.1. Technical Feasibility

Technical feasibility focuses on whether the chosen technology and methodology can meet the required performance of the system:

- **Antenna Design:** The transmission line model for rectangular microstrip patches is well-established, and MATLAB Antenna Toolbox provides validated `patchMicrostrip()` and `pifa()` objects for simulation. FR4 substrate parameters ($\epsilon_r = 4.4$, $h = 1.6$ mm) are known and consistently available locally.
- **Fabrication:** Toner-transfer chemical etching is a proven low-cost PCB fabrication technique that achieves dimensional tolerances of ± 0.3 mm — sufficient for 2.4 GHz patch dimensions of approximately 28–30 mm.
- **Measurement:** VNA access is provided by an industry partner at no cost. The two-antenna RSSI radiation pattern measurement method has been validated in IEEE-published literature as a low-cost alternative to an anechoic chamber.
- **Positioning System:** ESP32 and LoRa/Wi-Fi RSSI trilateration with Kalman filtering is well-documented in recent IEEE literature with demonstrated mean errors of 2.3 m for built-in antennas, providing a clear baseline for improvement.

2.5.2. Operational Feasibility

Operational feasibility evaluates how system can be used and maintained in real-world conditions:

- **User Simplicity:** The MATLAB base station provides a live position plot requiring no specialist operation. The ESP32 firmware runs autonomously once flashed, with no ongoing configuration required.

- **Minimal Maintenance:** The system has no moving parts, no external servers, and no internet dependency. Long-term deployment requires only battery replacement on the mobile node.
- **License-Free Operation:** ESP-NOW and Wi-Fi operate in the 2.4 GHz ISM band, which is unlicensed in Nepal, eliminating the need for regulatory approvals.

2.6. Economic Feasibility

Economic feasibility determines whether the system can be implemented at a reasonable cost:

- **Low Initial Cost:** Total hardware cost is below NPR 10,000. All components — ESP32 modules, FR4 copper-clad, SMA connectors, and etching chemicals — are available in Kathmandu and Pokhara without import.
- **VNA Access at No Cost:** VNA measurement, which would otherwise cost NPR 50,000–200,000 for equipment purchase, is provided by an industry partner, making the electrical validation free of charge.
- **Software:** MATLAB Antenna Toolbox is available through the university's academic license. No paid software beyond the university subscription is required.

3. Related Theory

3.1. Microstrip Patch Antenna

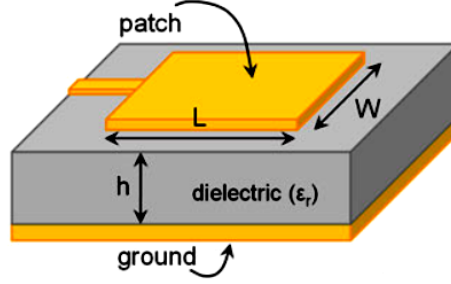


Figure 3.1: Microstrip Patch Antenna Geometry.

A microstrip patch antenna consists of a flat metallic radiating patch mounted on a grounded dielectric substrate. The rectangular patch geometry — the most widely used configuration — is designed using the transmission line model, which treats the patch as two radiating slots separated by a low-impedance transmission line. The patch width W and physical length L are calculated from the substrate's relative permittivity (ϵ_r), thickness (h), and the desired resonant frequency (f_0). An inset feed line at position y_0 from the patch edge provides $50\ \Omega$ impedance matching without the need for an external matching network.

The key parameters are: patch width W , which controls impedance; effective dielectric constant $\epsilon_{r,\text{eff}}$, which accounts for fringing fields; effective patch length L_{eff} and physical length L , corrected for end-effects via ΔL ; and feed position y_0 computed for $50\ \Omega$ matching. The standard rectangular patch achieves approximately 5-7 dBi gain, a known broadside radiation pattern, and a -10 dB fractional bandwidth of 1-4% on FR4 at 2.4 GHz. In this project, the rectangular patch serves as the Stage 1 baseline, while compact variants — meandered, defected ground structure (DGS), and shorted patch (PIFA-type) geometries — are explored in Stage 2 to reduce physical footprint while maintaining $S_{11} \leq -10$ dB at 2.4 GHz.

3.2. FR4 Substrate



Figure 3.2: FR4 Substrate.

FR4 is a glass-reinforced epoxy laminate material widely used in printed circuit boards (PCBs). It consists of woven fiberglass cloth impregnated with epoxy resin, providing a balance of electrical performance, mechanical stability, and low cost. For antenna applications, FR4 is characterized by its relative permittivity ϵ_r and loss tangent $\tan \delta$, which vary with frequency and temperature. At 2.4 GHz, typical FR4 properties include $\epsilon_r \approx 4.4$ and $\tan \delta \approx 0.02$, making it suitable for low-to-moderate frequency applications but introducing measurable losses that affect antenna efficiency.

The substrate thickness h is a critical design parameter, influencing impedance matching, bandwidth, and radiation characteristics. Common FR4 thicknesses range from 0.8 mm to 1.6 mm. Thinner substrates (e.g., 0.8 mm) reduce fringing fields and improve bandwidth but require more precise fabrication. Thicker substrates (e.g., 1.6 mm) simplify etching and reduce coupling between layers but increase substrate losses and reduce antenna efficiency. In this project, FR4 serves as the substrate for all antenna designs, with thickness optimized to balance bandwidth, size, and performance requirements.

3.3. ESP32 Microcontroller

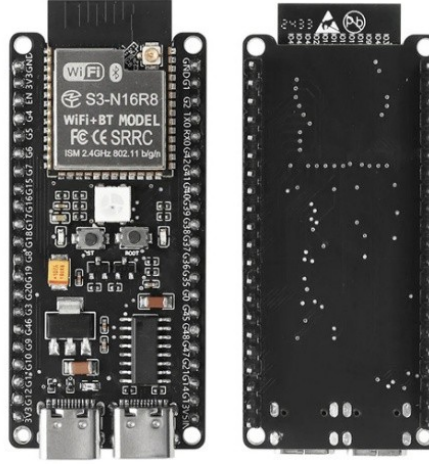


Figure 3.3: ESP32 Microcontroller.

ESP32 is a low-cost, low-power system-on-chip (SoC) microcontroller developed by Espressif Systems. It features a dual-core 240 MHz processor, 520 KB SRAM, built-in Wi-Fi (802.11 b/g/n at 2.4 GHz), and Bluetooth (Classic and BLE). In this project, the ESP32 serves multiple roles: the mobile node broadcasts ESP-NOW beacons; anchor nodes receive beacons and measure RSSI; and the gateway ESP32 aggregates RSSI data from all three anchors and forwards the combined packet to the laptop over USB-serial at 115200 baud. The ESP-NOW protocol, which is a connectionless peer-to-peer Wi-Fi extension developed by Espressif, allows all three anchors to receive a single broadcast packet simultaneously in under 10 ms — a latency advantage over conventional Wi-Fi RSSI scanning, which requires sequential AP association at 200–500 ms per anchor. ESP32 development boards are readily available in Kathmandu and Pokhara at approximately NPR 1000 each.

3.4. Power Supply Design

Each node uses a power source matched to its role. The mobile node, being portable, is powered by a single 18650 lithium-ion cell (3.7 V, 2600 mAh) managed by a TP4056 charging module with a MT3608 boost converter providing 5 V to the ESP32 VIN pin, giving an estimated runtime of 8–12 hours. The three anchor nodes are fixed and powered from standard 5 V USB wall adapters, ensuring continuous stable operation without battery management. The gateway node draws power directly from the laptop USB port through the existing serial connection, requiring no separate supply.

Table 3.1: Power Supply Configuration

Node	Power Source	Runtime
Mobile node	18650 Li-ion + TP4056 + MT3608	8–12 hours
Anchor A, B, C	5 V USB wall adapter	Continuous
Gateway	Laptop USB	Continuous

3.5. Vector Network Analyzer (VNA)



Figure 3.4: Vector Network Analyzer(VNA)

A Vector Network Analyzer (VNA) is an instrument that measures the S-parameters of RF components and antennas over a specified frequency range. For a single-port antenna, the primary measurement is S_{11} (reflection coefficient), which describes how much power is reflected from the antenna back into the feed line. An S_{11} value below -10 dB at the desired frequency indicates that more than 90% of the applied power is accepted by the antenna (i.e. not reflected), and the corresponding frequency range defines the -10 dB impedance bandwidth. The VNA also displays the antenna impedance on a Smith chart, allowing the feed position y_0 to be adjusted for minimum reflection. In this project, VNA access is provided by an industry partner at no cost, and a one-port SOL calibration will be performed at the SMA reference plane before measurement. If a two-port VNA is available, an S_{21} measurement between two identical patch antennas will be used with the Friis transmission equation to estimate realized gain.

3.6. Matlab Tool Box

The MATLAB Antenna Toolbox provides a library of analytically modeled and numerically simulated antenna objects, including `patchMicrostrip()` for rectangular microstrip patches

and pifa() for planar inverted-F antennas. The toolbox uses a method-of-moments solver to compute S-parameters, impedance, and radiation patterns in three dimensions. In this project, the toolbox is used to: simulate S_{11} over 1–4 GHz and confirm resonance at 2.4 GHz; perform parametric sweeps of feed position y_0 in 0.5 mm steps to find the global S_{11} minimum; extract E-plane and H-plane radiation patterns for comparison against measured RSSI-based patterns; and explore compact patch variants before committing to fabrication. The Antenna Designer app within the toolbox provides an interactive interface for initial parameter tuning before scripted batch sweeps.

3.7. Path Loss Model

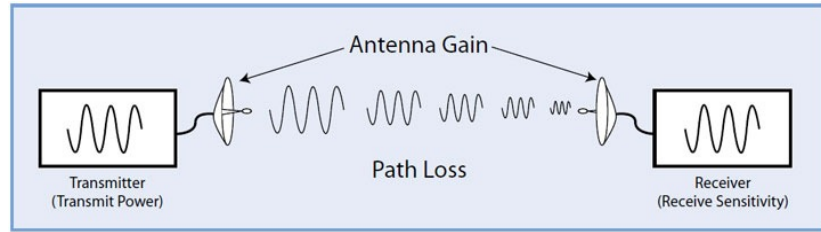


Figure 3.5: Path Loss model.

The path loss model provides the mathematical relationship between the received signal strength and the physical distance separating a transmitter and receiver. In this project, the log-distance path loss model is employed to convert the RSSI measurements recorded at each anchor node into distance estimates, which are subsequently used as inputs to the trilateration algorithm. The model is expressed as:

$$\text{RSSI}(d) = \text{RSSI}(d_0) - 10n \log_{10} \left(\frac{d}{d_0} \right) \quad (3.1)$$

where,

- $\text{RSSI}(d)$ is the received signal strength at distance d (in dBm),
- $\text{RSSI}(d_0)$ is the received signal strength at a reference distance d_0 (typically 1 meter),
- n is the path loss exponent, which depends on the propagation environment,
- d is the actual distance between the transmitter and receiver.

Rearranging for distance gives the inversion formula that would be used in firmware:

$$d = d_0 \cdot 10^{\frac{\text{RSSI}(d_0) - \text{RSSI}(d)}{10 \cdot n}} \quad (3.2)$$

The path loss exponent n is empirically determined for the specific testing environment, as follows: the mobile node is placed at different distances from a reference anchor, the average RSSI values are collected, and a linear regression line is plotted against $\log_{10}(d)$ for RSSI. In the case of the open indoor environment, n will have a value between 2.0 and 2.5, while for the cluttered indoor environment with obstacles, n will be within the range 2.5 to 3.5. The RSSI reference for $d_0 = 1$ m is obtained through an average of thirty RSSI samples in order to eliminate the short-term multipath effects. As it was previously mentioned, the log-distance model makes assumptions about spatially isotropic radiation from the mobile node, and thus, the performance of distance measurement depends on the isotropy of the radiation of the transmitting antenna. Almasi [4] reported that for asymmetric antennas, the distance error can be up to 1.5 meters due to the systematic effect on distance measurement, which is why replacing the onboard antenna with a custom-designed one became necessary for the project.

3.8. Trilateration

Trilateration is a geometric method used to determine the absolute or relative location of a point by measuring its distances to several known reference points (anchors). Unlike triangulation, which relies on the measurement of angles, trilateration relies exclusively on distance measurements (such as Time of Flight or Received Signal Strength Indicator in wireless networks). In a two-dimensional Cartesian plane, determining an unambiguous position requires distances to at least three non-collinear reference points, which corresponds geometrically to finding the single point of intersection of three distinct circles.

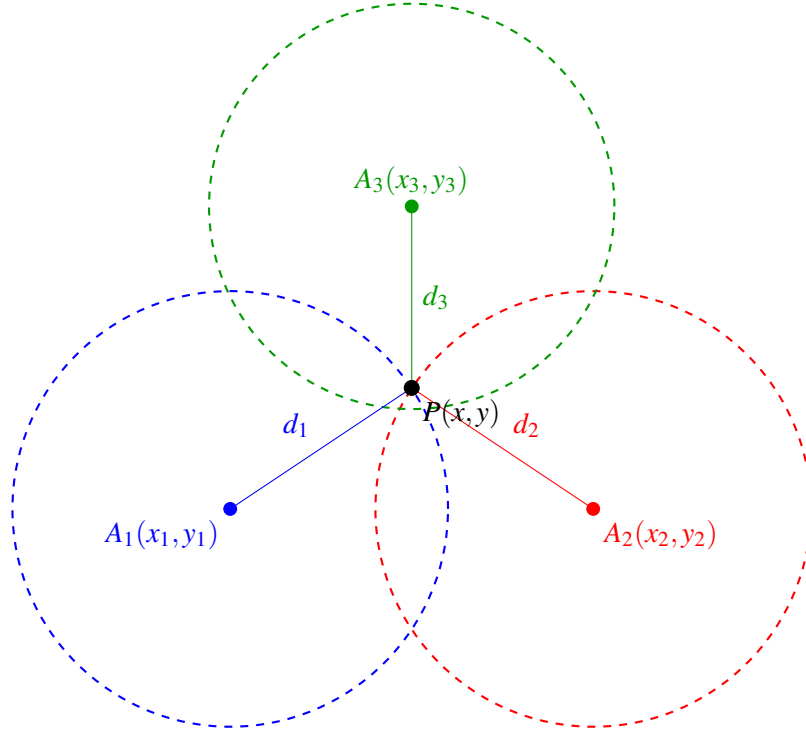


Figure 3.6: Trilateration: position of mobile node $P(x, y)$ estimated from distances d_1, d_2, d_3 to three anchor nodes.

3.9. Kalman Filter

The Kalman Filter is an optimal recursive data processing algorithm used to estimate the internal state of a linear dynamic system from a series of noisy measurements. It is heavily utilized in control systems, robotics, and sensor fusion (like combining GPS and IMU data). The algorithm operates in a continuous, two-step feedback loop: the Predict step, which projects the current state estimate and error covariance forward in time based on the system's physical model; and the Update step, which incorporates a new noisy measurement to correct the predicted state. The filter dynamically calculates a weighting factor, known as the Kalman Gain, which determines whether to trust the theoretical model or the actual measurement more heavily based on their respective uncertainties.

4. Literature Review

The domain of indoor positioning using radio frequency (RF) signals has been extensively studied, with a growing body of work addressing antenna design, system architecture, and localization algorithms. The following review surveys the key works directly relevant to this project.

The most directly comparable system baseline is provided by Mousa et al. [1], who implemented Wi-Fi RSSI trilateration for autonomous ground vehicle positioning using ESP32 modules and reported a mean positioning error of 2.3 metres with built-in antennas. This result establishes the primary performance benchmark that the present work seeks to improve through custom antenna design.

The relationship between antenna radiation characteristics and localization accuracy was established by Cidronali et al. [2], who showed that antennas with known, symmetric radiation patterns produce positioning errors up to 40% lower than generic chip antennas in comparable indoor environments. This finding provides the core motivation for replacing the ESP32's built-in trace antenna with a characterized custom patch. A subsequent study by the same group [3] evaluated a custom star-shaped microstrip patch antenna at 2.4 GHz, validating its radiation pattern using RSSI measurements at 10-degree angular increments and demonstrating good agreement with simulation results without requiring a dedicated anechoic chamber. This work establishes the two-antenna RSSI rotation technique as a legitimate IEEE-published validation method directly applicable to the present project.

The specific consequences of using an uncharacterized built-in antenna were quantified by Almasi [4], who found that improper ESP32 antenna orientation introduces systematic RSSI errors of up to 8 dBm, corresponding to approximately 1.5 metres of additional ranging error at typical indoor distances. This result precisely defines the problem that a properly mounted patch antenna with a symmetric, characterized radiation pattern is designed to address.

A comprehensive survey of relative localization methods for robot swarms by Tran et al. [5] reviewed RSSI, TDOA, AOA, and UWB approaches and concluded that Wi-Fi RSSI trilateration represents the most practical entry point for low-cost swarm navigation, while identifying UWB as the recommended upgrade path for sub-30 cm accuracy. This supports the present project's framing of RSSI trilateration as a zone-level positioning technology and informs the system's upgrade trajectory. Similarly, Patra et al. [6] reviewed GPS-independent localization techniques for disaster rescue scenarios and confirmed that RF-based position-

ing is the most viable solution in GPS-denied environments, directly supporting the Nepal earthquake rescue motivation of this work.

The theoretical foundations for compact antenna miniaturization are provided by Wong [7] and Balanis [8], who document meandered patch geometries, defected ground structures (DGS), and shorted patch (PIFA-type) configurations as established techniques capable of reducing antenna footprint by 30–60% relative to a standard rectangular patch while maintaining acceptable gain and impedance bandwidth at 2.4 GHz. These techniques are directly applied in the Stage 2 compact antenna exploration phase of this project.

Finally, Halperin et al. [9] demonstrated that Channel State Information (CSI) fingerprinting achieves Centi-metre level mean positioning errors in indoor environments by exploiting multipath channel profiles. This result establishes CSI fingerprinting as the target upgrade path beyond the present RSSI-based implementation, consistent with the system architecture’s deliberate provision for a future CSI upgrade without hardware modification.

5. System Architecture and Methodology

5.1. Methodology

This project is divided into four main phases: designing and simulating the antenna, fabricating and measuring it, building the positioning system, and finally validating the overall performance. Each phase feeds into the next, so problems caught early — especially during simulation — save significant time and material cost later.

5.1.1. Requirement Analysis

Before starting any design, we first defined what the antenna actually needs to achieve. The core requirement is that the antenna must resonate at 2.4 GHz with an $S_{11} \leq -10$ dB, meaning at least 90% of the input power is radiated rather than reflected. Beyond that, the radiation pattern must be as symmetric and broadside as possible, because asymmetric nulls are precisely what causes inconsistent RSSI readings at different node orientations. The feed must be matched to 50Ω to interface directly with the ESP32's RF output. Finally, the antenna must be fabricable on FR4 copper-clad board — the only substrate realistically available in Nepal — using toner-transfer etching without specialist equipment.

5.1.2. Antenna Design

5.1.3. Rectangular Microstrip Patch Baseline

The rectangular microstrip patch was chosen as the Stage 1 design because its dimensions are fully determined by closed-form transmission line model equations, making it straightforward to compute before any simulation. The patch width W is calculated first to set the input impedance, followed by the effective dielectric constant $\epsilon_{r,\text{eff}}$ to account for fringing fields, and then the physical length L corrected for the end-effect extension ΔL . The inset feed position y_0 is then swept to find the point where the input impedance at resonance equals 50Ω .

5.1.4. Compact Variants (Schedule Permitting)

If time allows after Stage 1 fabrication and measurement, three compact geometries will be explored in simulation: a meandered patch (which folds the current path to reduce the physical footprint), a defected ground structure (DGS) variant (which etches slots in the ground plane to slow the effective wave velocity), and a shorted patch (PIFA-type) configuration (which uses a shorting wall to halve the resonant length). All three techniques are simulated in MATLAB before any attempt at fabrication, since each requires iterative tuning.

5.1.5. Simulation

All antenna simulation is carried out in the MATLAB Antenna Toolbox. A `dielectric()` object is created with FR4 parameters ($\epsilon_r = 4.4$, $\tan \delta = 0.02$, $h = 1.6$ mm), and the antenna is instantiated as a `patchMicrostrip()` object. An `sparameters()` frequency sweep is run from 1 to 4 GHz to confirm resonance at 2.4 GHz and check bandwidth. A parametric sweep of the feed position y_0 is then performed in 0.5 mm steps to find the minimum S_{11} . E-plane and H-plane radiation patterns are exported for direct comparison with the measured patterns obtained later using the two-antenna RSSI rotation method.

5.1.6. Fabrication

Once simulation results confirm the design meets the $S_{11} \leq -10$ dB requirement, the antenna is printed on A4 paper at 1:1 scale using a laser printer and transferred to FR4 copper-clad board using the toner-transfer method. The board is etched in ferric chloride solution. To hedge against fabrication tolerances, three boards are prepared with patch lengths of $L = 28$, 29, and 30 mm — bracketing the simulated optimal length. After etching, each patch is measured with a vernier caliper to verify dimensional accuracy, and an SMA connector is soldered at the feed point.

5.1.7. Antenna Measurement

Electrical characterization is performed on a VNA provided by our industry partner at no cost. A one-port SOL (Short-Open-Load) calibration is performed at the SMA reference plane before each measurement session. The S_{11} is swept from 1 to 4 GHz and the resonant frequency, -10 dB bandwidth, and Smith chart impedance locus are recorded. The radiation pattern is measured using the two-antenna RSSI rotation method described by Cidronali et

al. [3]: the patch under test is rotated in 10-degree increments through 360° while RSSI is logged from a fixed reference antenna, separately for the E-plane and H-plane.

5.1.8. Positioning System Development

The positioning system uses five ESP32 modules. The mobile node continuously broadcasts ESP-NOW beacon packets containing its MAC address. Each of the three fixed anchor nodes receives these beacons and records the RSSI value. A dedicated gateway ESP32 collects the RSSI readings from all three anchors via ESP-NOW and forwards the aggregated packet to a laptop over a single USB-serial connection at 115200 baud. On the laptop, a MATLAB script reads the serial stream, applies the log-distance path loss model to convert each RSSI to a distance estimate, and solves the trilateration system using least-squares. A Kalman filter is applied to the position estimates to reduce noise, and the live position is plotted at approximately 10 Hz.

5.1.9. Testing and Validation

Performance is evaluated at twenty pre-marked test positions spread across the test environment. At each position, thirty RSSI readings are averaged per anchor before trilateration is computed, to reduce short-term multipath fluctuation. The experiment is run twice for each test position: once with the custom patch antenna mounted on the mobile node, and once with the ESP32's built-in trace antenna, so that the improvement from the custom antenna is directly quantified under identical conditions. Ground truth distances are measured with a tape measure. Results are reported as a position error table, a Cumulative Distribution Function (CDF) graph showing error percentiles, and a statistical summary including mean error, median error, and the 90th percentile error for both antenna configurations.

5.2. Block Diagram

The complete system architecture is shown in Fig 5.1. The custom compact patch antenna is mounted exclusively on the mobile node. All three anchor nodes use the standard built-in ESP32 PCB trace antenna. Here, all the wireless tasks (represented by dotted line in below Fig 5.1) are performed using ESP-NOW protocol.

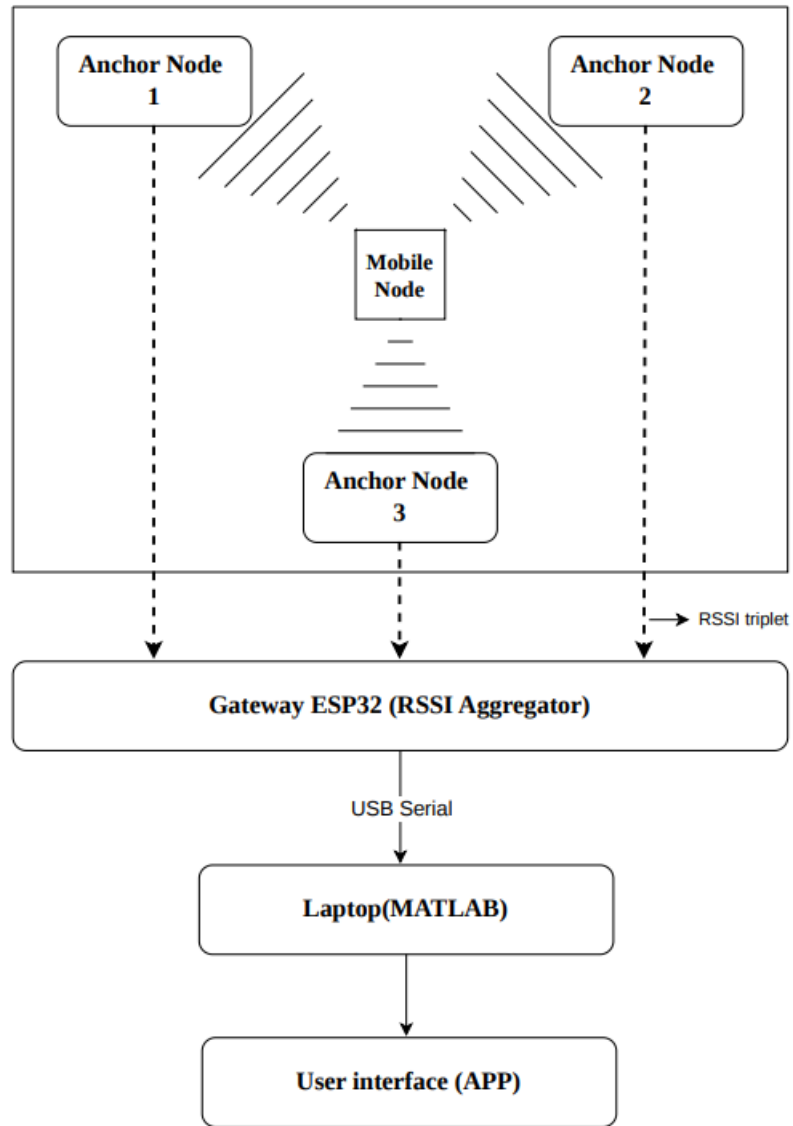


Figure 5.1: Block Diagram of the System

5.3. Circuit/System Diagram

The custom patch antenna is connected to the mobile node ESP32 via a U.FL-to-SMA pig-tail cable, with the SMA female connector soldered to the patch feed point. The hardware connections and module interfacings are shown in Figure 5.2.

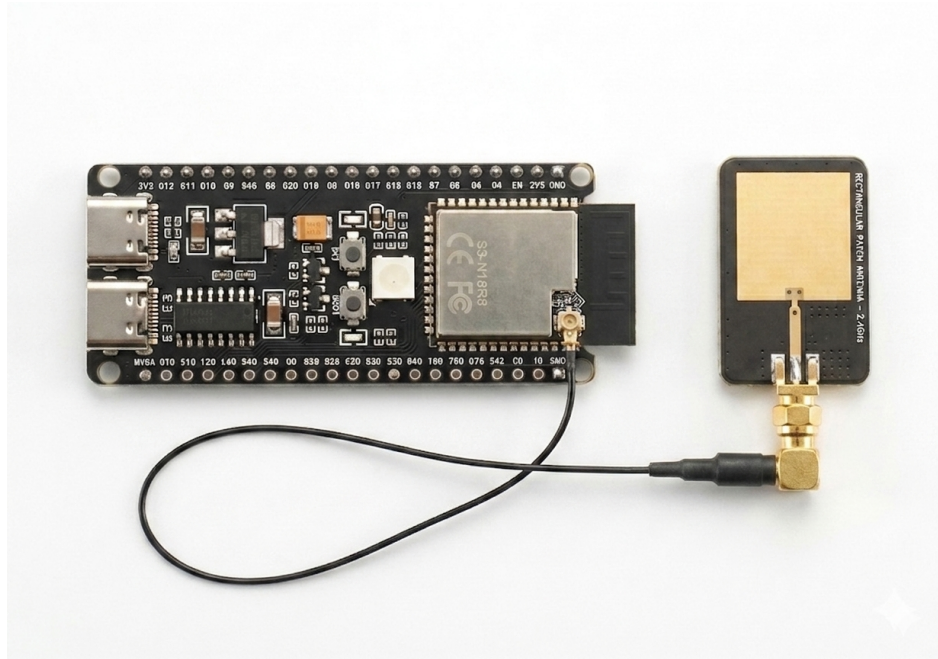


Figure 5.2: Circuit/System Diagram.

5.4. Flowchart

The firmware flowchart describes the ESP-NOW messaging loop, as shown in Figure 5.3. On power-on, the modules are initialized and ESP-NOW is configured. The mobile node continuously broadcasts a beacon. Each anchor node, upon receiving a beacon, measures the RSSI and forwards the value along with its anchor ID to the gateway via an ESP-NOW unicast message. The gateway assembles the RSSI triplet and sends it over USB-serial to MATLAB. MATLAB applies the path loss model, runs the trilateration algorithm, applies the Kalman filter, and updates the live position plot.

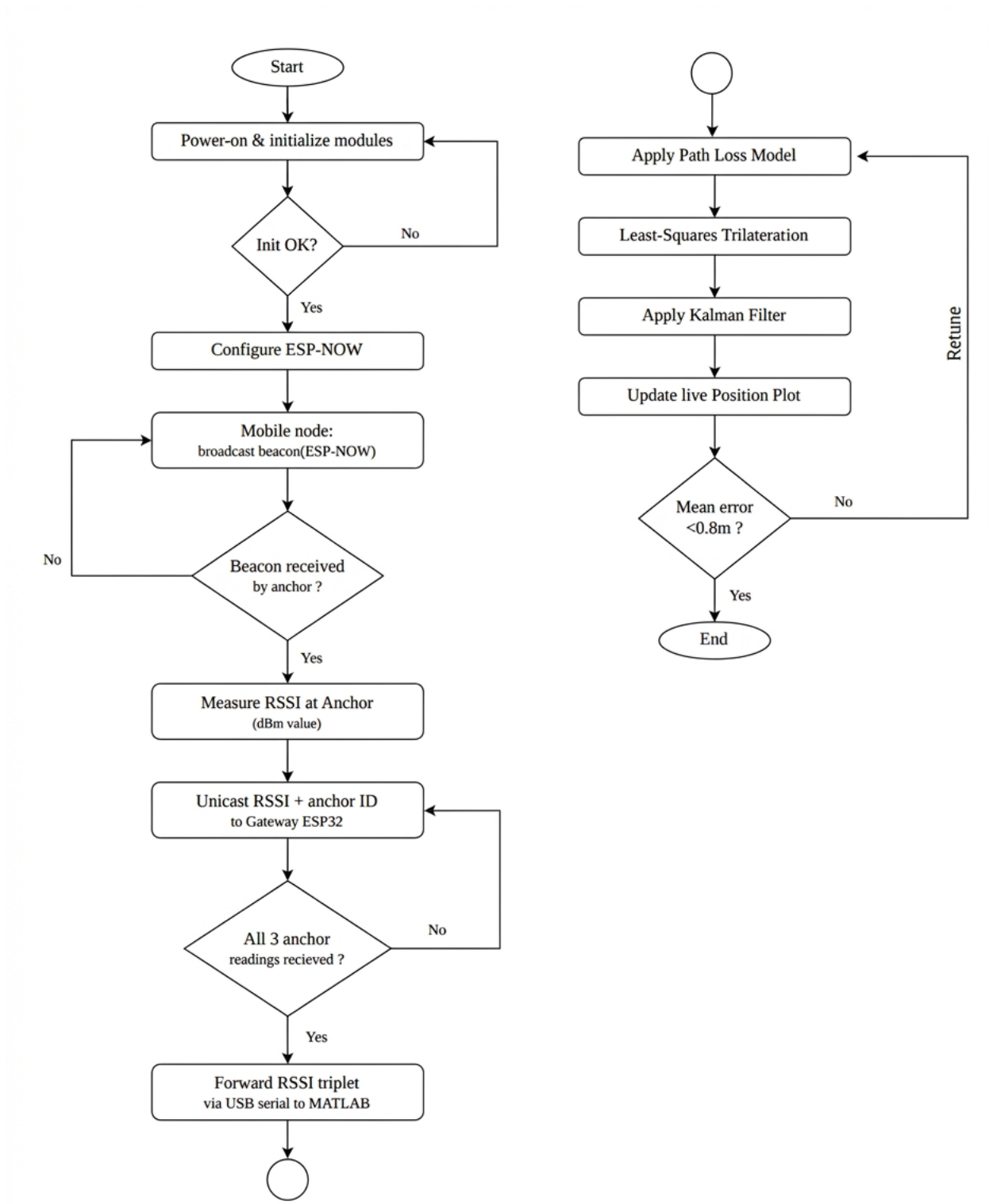


Figure 5.3: Flowchart of the Proposed System.

6. Epilogue

6.1. Expected Output

6.1.1. Custom Antenna Design and Characterization

A validated MATLAB script computing patch antenna dimensions (W, L, y_0) with parametric sweep results from MATLAB Antenna Toolbox, achieving $S_{11} \leq -10$ dB at 2.4 GHz in simulation. If a compact variant achieves comparable performance, its dimensions, footprint reduction, and simulated efficiency will also be reported.

6.1.2. VNA Measurement and Fabrication Validation

A fabricated compact patch antenna on FR4 with VNA-measured S_{11} resonance dip, -10 dB bandwidth, complex impedance $(R + jX)$, and gain estimate from the S_{21} /Friis method. Measured E-plane and H-plane radiation patterns will be compared against MATLAB simulation with quantified agreement.

6.1.3. Positioning Accuracy Improvement

Positioning accuracy improvement of at least 40% — from approximately 2.0 m mean error with the built-in antenna to below 1.2 m with the custom patch antenna — demonstrated across twenty test positions and presented as a Cumulative Distribution Function (CDF) comparison graph.

6.1.4. Low-Cost Reproducible System

A complete, reproducible, low-cost indoor positioning system using five ESP32 modules and locally available components, with a total hardware cost below NPR 10,000. A characterized, documented antenna design reusable by subsequent student projects.

6.2. Budget Analysis

The complete estimated cost of the system components is summarized in Table 6.1.

Table 6.1: Cost of System Components.

S.N.	Component	Qty	Unit Cost (NPR)	Total (NPR)
1	ESP32 S3-N16R8 Development Board	4	1,400	5,600
2	ESP-32S Development Board	1	700	700
3	FR4 Copper Clad Board (30×20 cm)	1	700	700
4	SMA Male Connector	5	250	1,250
5	U.FL to SMA Female Adapter	5	150	750
6	Battery	2	100	200
7	Miscellaneous	-	500	500
TOTAL ESTIMATED BUDGET				9,700

6.3. Work Schedule

The timeline for the project development is outlined below, and visually represented in the Gantt chart in Figure 6.1.

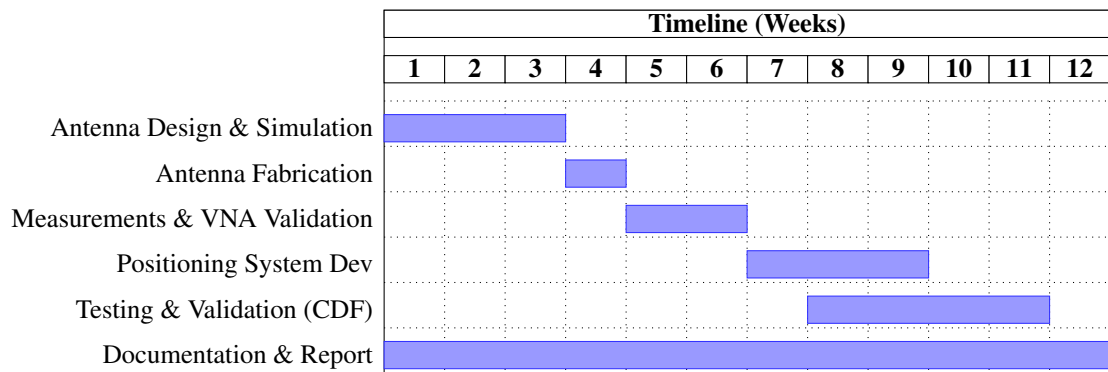


Figure 6.1: Project Gantt Chart detailing the 12-week schedule.

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